

Exploratory Data Analysis (EDA) Report

Project: Data Analytics Project 2

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn

Project Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the provided dataset to identify trends, distributions, customer behavior, outliers, and meaningful business insights.

Steps Performed

- Imported required Python libraries.
- Loaded Excel dataset using Pandas.
- Inspected dataset using head(), tail(), sample(), info(), shape(), and columns.
- Performed data cleaning and handled missing values in CouponCode column.
- Checked and removed duplicate values.
- Calculated descriptive statistics such as mean, median, minimum, maximum, and total sales.
- Performed Product Analysis using value_counts().
- Analyzed Order Status and Payment Methods.
- Performed Referral Source Analysis.
- Converted Date column into datetime format for time analysis.
- Created monthly sales trends using line charts.
- Performed distribution analysis using histogram.
- Detected outliers using boxplot and IQR method.
- Performed correlation analysis using heatmap.
- Generated business insights and final conclusion.

Key Findings

1. Most orders were completed successfully.
2. Some products generated higher sales compared to others.
3. Payment methods such as UPI/Credit Card were commonly used.
4. Monthly sales showed fluctuations over time.
5. Outliers were identified in TotalPrice column.
6. Correlation analysis helped understand relationships between numerical columns.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

- Jupyter Notebook

Conclusion

This Exploratory Data Analysis project successfully analyzed the dataset and identified important business insights, trends, customer behavior, distributions, and outliers. The project demonstrates practical knowledge of data cleaning, data visualization, statistical analysis, and analytical thinking.